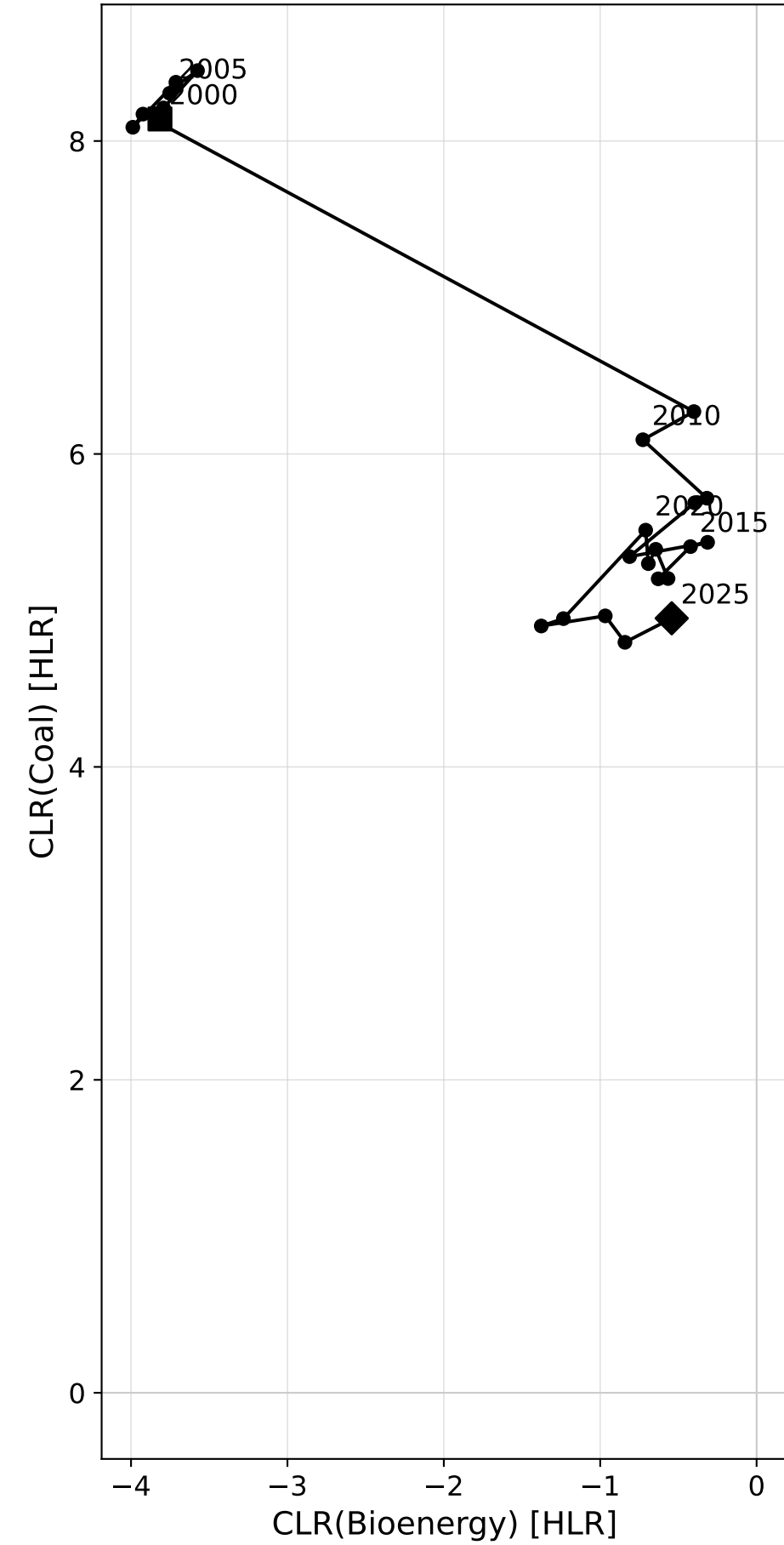
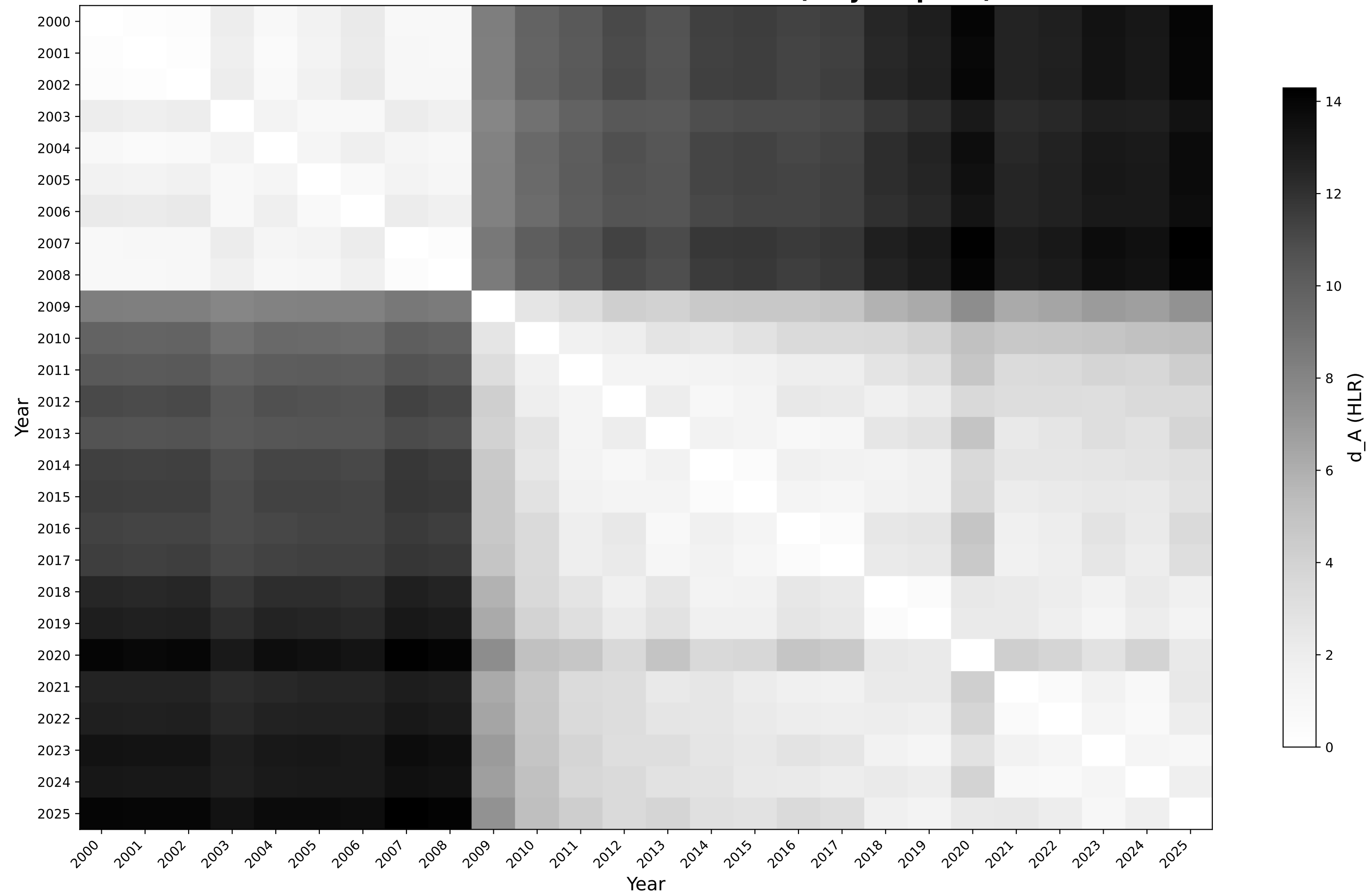


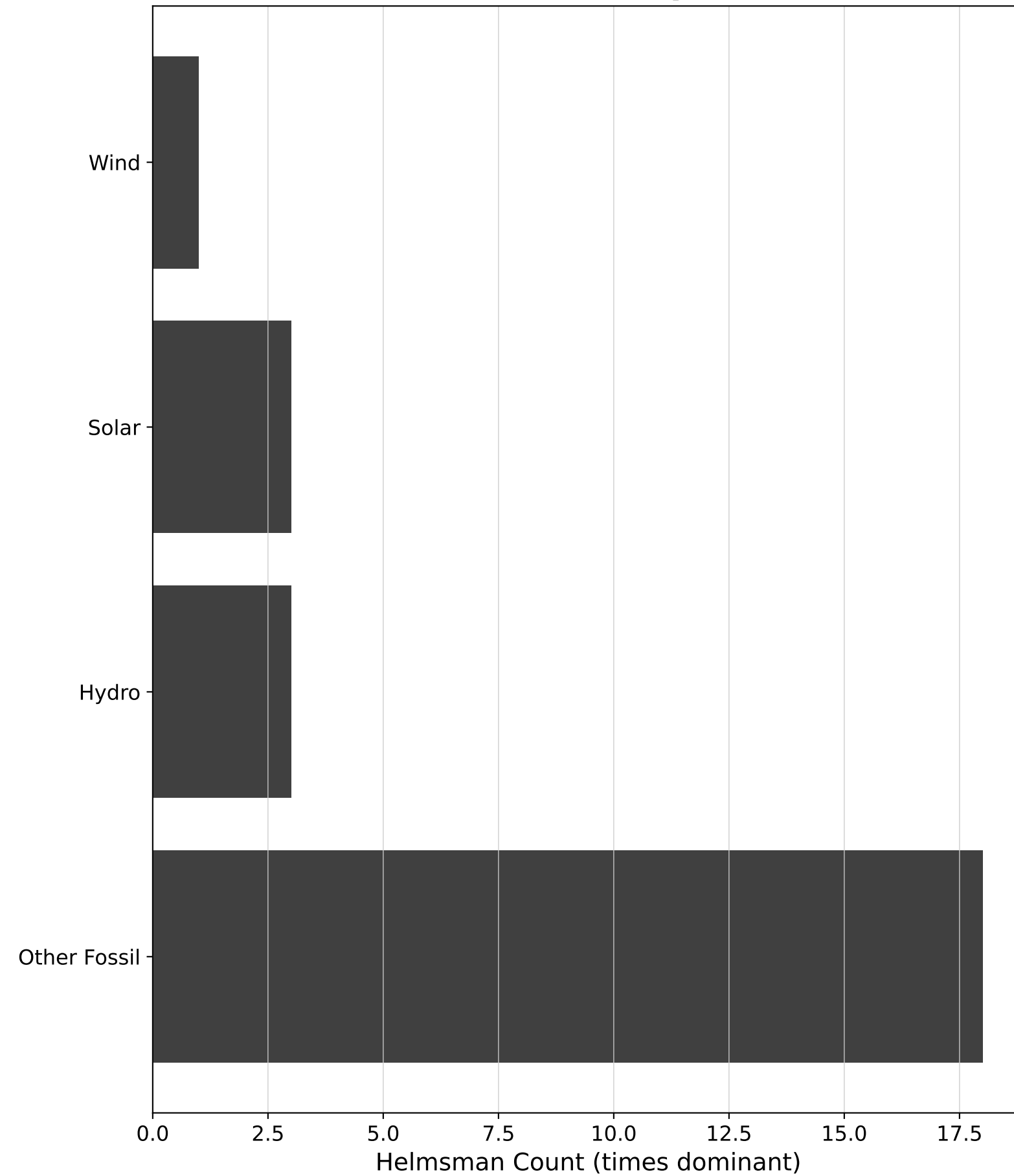
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



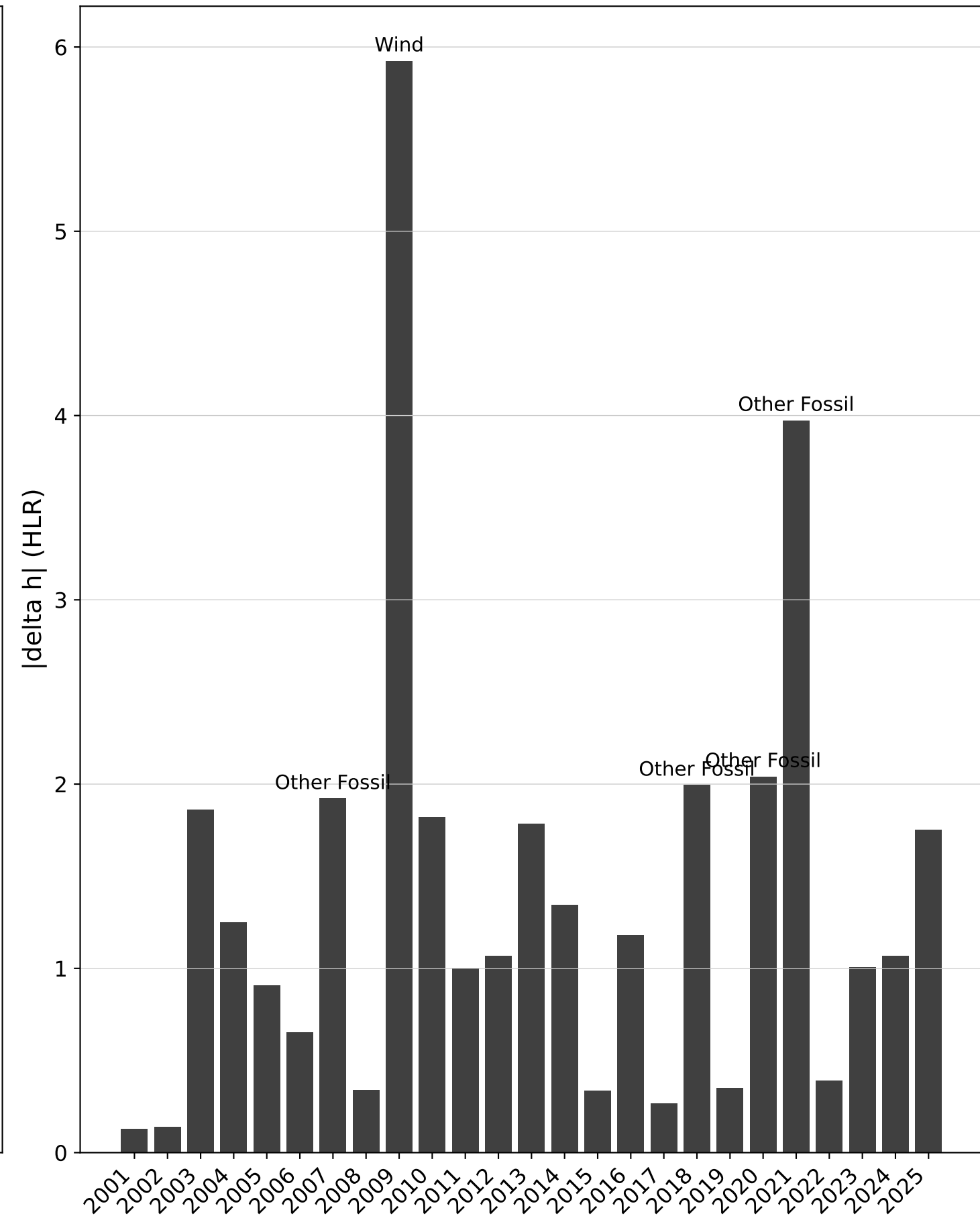
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



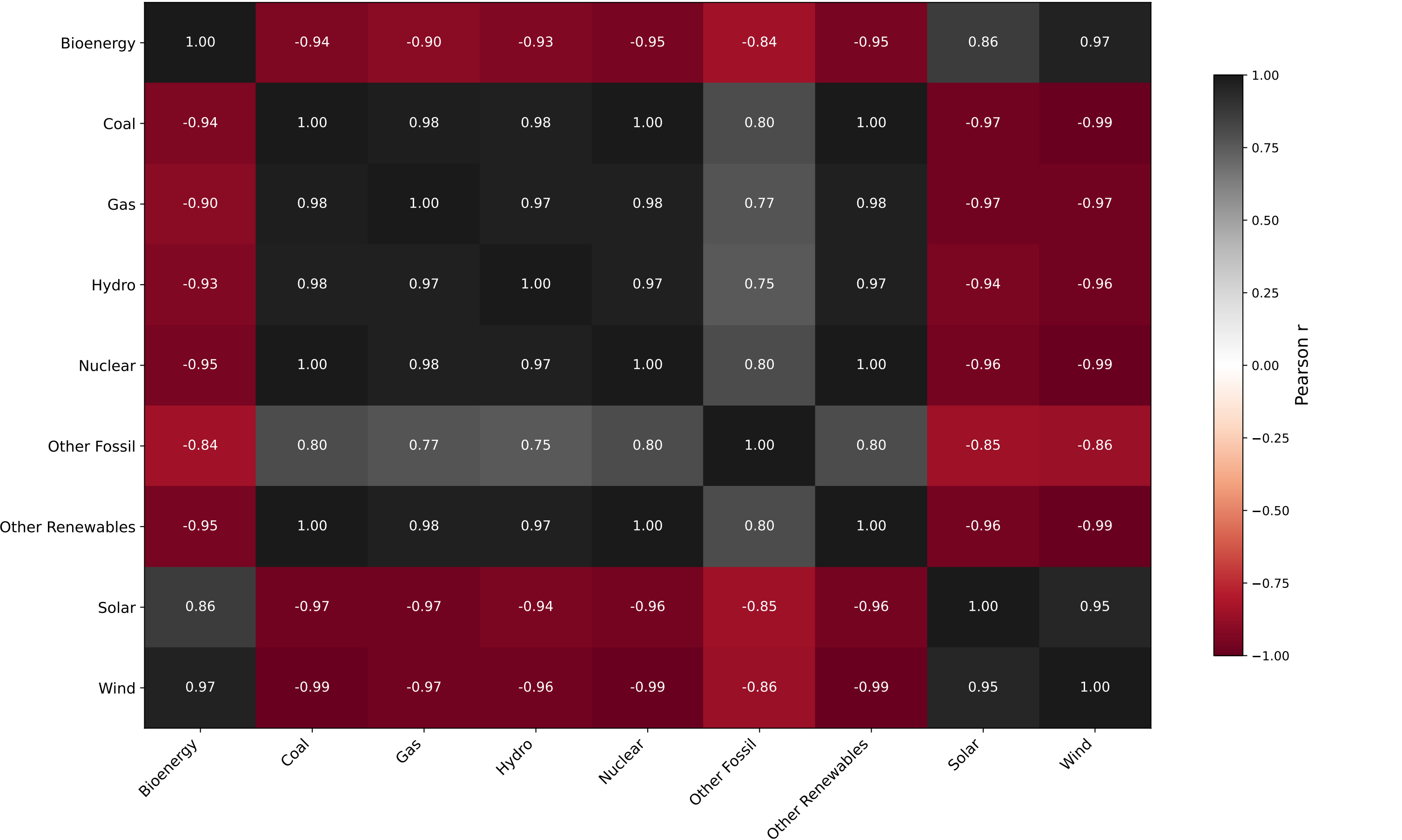
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2007	2025	14.2906	61.2	Solar
2	2007	2020	14.2291	59.4	Solar
3	2008	2025	14.1193	60.6	Solar
4	2008	2020	14.0101	58.6	Solar
5	2000	2025	13.9781	61.2	Solar
6	2000	2020	13.9643	59.5	Solar
7	2002	2025	13.9500	61.3	Solar
8	2002	2020	13.9430	59.6	Solar
9	2001	2025	13.9193	61.0	Solar
10	2001	2020	13.8894	59.2	Solar
11	2004	2025	13.7119	60.1	Solar
12	2005	2025	13.7028	59.0	Solar
13	2007	2023	13.6669	58.8	Solar
14	2004	2020	13.6104	58.0	Solar
15	2006	2025	13.6028	58.1	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2006	2009	2020	28.4334	1.75
2	2005	2009	2020	27.6942	1.77
3	2007	2009	2020	27.5941	1.88
4	2008	2009	2020	27.5393	1.85
5	2003	2009	2020	27.4006	1.72
6	2007	2020	2021	26.8451	3.36
7	2008	2020	2021	26.7023	3.31
8	2007	2020	2024	26.6745	3.59
9	2004	2009	2020	26.6131	1.79
10	2000	2009	2020	26.5906	1.84

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.8255	Hs-5	193.2	0.00	0.0000	---	172235
2001	0.8319	Hs-5	192.4	0.81	0.1976	Hydro	174665
2002	0.8421	Hs-5	191.5	0.80	0.1952	Hydro	178405
2003	0.8437	Hs-5	194.1	8.29	2.0096	Other Fossil	181915
2004	0.8534	Hs-5	191.1	5.75	1.3954	Other Fossil	187875
2005	0.7957	Hs-5	202.9	4.49	1.1771	Hydro	195875
2006	0.7801	Hs-5	208.4	2.76	0.7157	Other Fossil	204985
2007	0.7553	Hs-5	209.6	8.26	2.0833	Other Fossil	207835
2008	0.7717	Hs-5	207.5	1.49	0.3837	Other Fossil	208385
2009	0.7253	Hs-4	148.4	36.02	8.4868	Wind	208382
2010	0.6690	Hs-4	158.3	11.90	2.5981	Solar	208922
2011	0.6594	Hs-4	150.6	7.25	1.6025	Solar	206562
2012	0.6208	Hs-4	161.1	5.62	1.2938	Other Fossil	202722
2013	0.5877	Hs-4	150.9	8.78	1.9545	Other Fossil	199292
2014	0.5752	Hs-4	161.4	6.88	1.5568	Other Fossil	214382
2015	0.5817	Hs-4	159.5	2.19	0.4902	Other Fossil	218182
2016	0.5621	Hs-4	154.5	5.87	1.2992	Other Fossil	221562
2017	0.5547	Hs-4	155.9	2.14	0.4680	Other Fossil	220892
2018	0.5256	Hs-4	173.7	9.37	2.2081	Other Fossil	223262
2019	0.4921	Hs-3	176.2	1.98	0.4671	Solar	224862
2020	0.4696	Hs-3	202.4	8.30	2.2050	Other Fossil	222553
2021	0.4466	Hs-3	164.7	17.01	4.2325	Other Fossil	224142
2022	0.4128	Hs-3	169.9	2.10	0.5141	Other Fossil	228882
2023	0.4098	Hs-3	176.3	4.90	1.1513	Other Fossil	230252
2024	0.4080	Hs-3	168.6	4.98	1.1796	Other Fossil	237522
2025	0.3988	Hs-3	183.3	7.76	1.8778	Other Fossil	242222